

AKSHIT SHARMA

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SUMMARY

Final-year CS engineering student building production-grade ML systems end to end — from custom C++ Chromium instrumentation to multi-agent RAG pipelines. Shipped Headless_BAI, an industrial pipeline combining DOM-level binary instrumentation, K-Means behavioral clustering, and LLM-based layout routing, achieving a 31.42% reduction in UI spatial friction across 1,000+ sessions. Top 5% at PDPM IITDM Deep Learning Summer School; Google Student Ambassador.

TECHNICAL SKILLS

Programming: Python, C++ (Chromium internals), TypeScript, SQL, Go

ML/AI: TensorFlow/Keras, PyTorch, scikit-learn, LSTM, NLP, K-Means clustering, LLM routing & inference optimization, model serving

GenAI & Agentic Systems: LangGraph multi-agent orchestration, RAG, Qdrant (vector DB), prompt engineering, fine-tuning (QLoRA)

Software Engineering & Systems: React/TypeScript, FastAPI, Flask, REST API design, Object-Oriented Design, n8n automation

MLOps & Cloud: Docker, GCP Cloud Run, AWS, IBM Cloud, CI/CD, Git

Data: Supabase, MySQL, pandas, NumPy

EXPERIENCE

AI/ML Engineering Intern — Future Finders *Jan – Jun 2026*

- Built and validated deep learning models end-to-end — data preprocessing, training, and performance evaluation — for production problem statements in a professional ML environment
- Engineered reusable model training pipelines and validation frameworks to standardize experimentation across projects

Deep Learning Intensive — PDPM IITDM Jabalpur *Jun 2025*

- Completed an intensive deep learning curriculum covering advanced neural network architectures, training methodologies, and real-world deployment strategies; ranked top 5% of 100+ participants (Excellence Award)

Cloud ML Deployment — IBM Cloud Trainee *Sep – Oct 2024*

- Deployed containerized ML applications with auto-scaling policies on IBM Cloud, gaining hands-on production infrastructure exposure

KEY PROJECTS

Headless_BAI – AI Layout Optimization Engine *Jan 2026 – Present*

- Custom C++ Chromium binary extracts DOM geometry via native memory interception; <100ms rehydration with zero CLS
- Llama-3-8B SLM generates deterministic JSON layout patches in <500ms (T=0.1 inference); 3 behavioral cohorts identified via K-Means (WCSS: 23.24)
- End-to-end telemetry pipeline (sensor.js → FastAPI → Supabase) processing 10,800+ behavioral events in real time
- Achieved 31.42% session-weighted reduction in UI spatial friction, validated across 1,000+ user sessions

GeoSmart Advisor – Geospatial Business Site Selection Intelligence System *Jun – Sep 2025*

- Engineered a cloud-agnostic intelligence system using a locally-hosted, 4-bit quantized Phi-3-mini SLM for narrative report generation (<900ms latency)
- Developed a Julia computational core (1.10) for spatial matrix operations, achieving a 15–40x speedup over Python (<180ms) while capping peak RAM at 420MB
- Implemented a React/deck.gl (WebGL) frontend with a custom asymmetric loss function (1.5x penalty for overconfidence) to ensure conservative, defensible site recommendations

DeepResearch AI Agent – Multi-Agent RAG System*Feb – Mar 2026*

- Designed and built an agentic RAG system using LangGraph for multi-agent orchestration (research, analysis, and memory agents)
- Integrated Qdrant for high-performance semantic search against document embeddings, reducing hallucinations and improving context retrieval
- Deployed via Python/FastAPI, Docker, and CI/CD on GCP Cloud Run, achieving <900ms end-to-end inference latency

HONORS

Excellence Award, Deep Learning Summer School — PDPM IIITDM Jabalpur (top 5% of 100+ participants) | Jun 2025
Google Student Ambassador — Leadership & community impact in technology | Dec 2025

EDUCATION

B.Tech Computer Science Engineering — Bahra University | Expected 2026

CERTIFICATIONS

AWS Educate ML Foundations (Jun 2025) | Cyber Security – IBM Certification Program (Mar 2026) | Geo Processing with Python – ISRO (Jun 2023) | Full Stack Web Development – Amazon/Coursera (Feb 2025)